

HAMER, Norway

WMO: 013850

Lat: 60.818N

Lon: 11.069E

Elev: 729

StdP: 14.31

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	2.6	-5.2	4.3	-1.2	-1.2	5.3	3.0	17.6	32.0	14.8	32.7	2.2	60	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	79.7	60.6	75.6	60.1	71.9	58.6	63.9	73.0	62.4	70.5	60.7	68.4	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.0	82.3	66.2	59.3	77.3	64.7	57.7	73.0	63.5	29.5	73.3	28.4	70.2	27.2	68.6	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.0	11.1	-3.6	83.9	5.5	3.1	-7.6	86.1	-10.9	87.9	-14.0	89.6	-18.0	91.8		
(5)			-5.4	66.5	5.9	2.6	-9.6	68.3	-13.1	69.8	-16.4	71.3	-20.7	73.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	21.8	23.5	31.7	41.3	50.9	57.8	62.2	58.7	51.8	41.0	32.2	24.8		
	DBStd	15.80	9.55	9.54	7.77	6.44	7.55	5.77	5.47	4.16	5.02	6.10	7.20	10.20		
	HDD50	4176	874	743	566	270	80	4	0	1	37	285	535	781		
	HDD65	8609	1339	1163	1031	710	443	230	125	199	396	743	984	1246		
	CDD50	1104	0	0	0	10	107	237	380	271	92	7	0	0		
	CDD65	61	0	0	0	0	4	13	40	4	0	0	0	0		
	CDH74	484	0	0	0	0	43	123	306	12	0	0	0	0		
	CDH80	74	0	0	0	0	2	13	59	0	0	0	0	0		
	WSAvg	4.7	4.3	3.9	4.9	5.1	5.5	5.4	4.8	4.5	4.8	4.6	4.5	4.1		
	PrecAvg	20.8	1.1	0.9	1.0	1.0	1.5	2.3	2.9	2.7	2.4	2.0	1.7	1.2		
(15) Precipitation	PrecMax	28.5	2.8	2.8	2.8	2.6	3.0	5.6	6.5	5.7	4.6	3.8	3.0	2.8		
	PrecMin	14.8	0.0	0.1	0.0	0.0	0.3	0.2	0.9	0.4	0.6	0.3	0.5	0.2		
	PrecStd	3.6	0.7	0.6	0.6	0.7	0.8	1.4	1.2	1.5	1.1	1.1	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	43.4	53.7	65.5	78.7	81.9	85.0	75.7	69.2	60.2	51.0	45.7		
		MCWB	39.1	37.5	43.6	50.5	57.1	60.3	64.0	61.8	56.9	55.1	47.4	42.0		
	2%	DB	38.7	39.9	49.2	61.3	73.6	78.0	81.4	72.9	65.9	55.6	47.2	42.9		
		MCWB	36.0	36.1	40.9	48.6	56.3	58.5	62.7	60.1	55.7	49.4	44.1	40.2		
	5%	DB	36.0	37.4	45.3	57.1	68.8	73.0	77.4	69.9	63.0	52.8	44.5	40.2		
		MCWB	33.8	34.5	38.3	46.6	54.7	57.6	61.5	58.7	55.1	48.5	42.2	37.4		
	10%	DB	33.8	35.1	41.7	52.8	64.2	68.4	73.1	67.0	60.2	50.1	41.5	37.3		
		MCWB	32.1	33.1	36.3	44.0	52.3	55.3	60.6	57.6	53.7	46.2	39.6	35.0		
	0.4%	WB	40.0	39.1	44.5	52.4	61.2	62.8	67.5	64.8	60.8	55.9	48.1	42.8		
		MCDB	42.1	41.5	52.4	62.0	69.5	70.7	77.8	70.7	63.0	59.5	50.6	44.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	36.8	41.4	49.6	58.6	61.0	64.8	63.1	58.4	51.6	44.8	40.4		
		MCDB	38.5	39.2	48.1	57.7	69.4	71.7	75.9	68.8	62.8	54.3	46.5	42.4		
	5%	WB	34.2	34.9	39.2	47.5	55.9	59.5	63.4	61.3	56.4	49.2	42.4	37.9		
		MCDB	35.7	37.0	43.6	56.1	65.3	69.8	73.5	66.7	61.5	51.8	44.2	40.0		
	10%	WB	32.5	33.4	37.0	45.0	53.6	57.6	61.7	59.6	54.6	46.8	39.9	35.3		
		MCDB	33.8	35.2	41.1	52.5	62.1	66.7	70.7	64.4	59.1	49.4	41.3	37.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.4	12.9	15.8	16.0	15.8	15.8	14.4	13.6	10.2	7.7	8.6		
		MCDBR	9.5	11.7	17.1	22.2	22.8	21.5	21.5	18.5	17.7	12.5	11.2	11.4		
		MCWBR	8.2	8.4	11.0	12.1	10.1	9.7	8.3	9.0	10.2	9.2	9.1	9.7		
		MCDBR	9.3	9.9	16.2	20.4	19.1	18.2	18.6	14.2	15.2	10.9	10.3	10.9		
	5% WB	MCWBR	8.2	7.6	10.9	11.3	9.2	9.1	8.2	8.1	9.5	9.0	9.0	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.277	0.306	0.346	0.349	0.343	0.361	0.352	0.321	0.306	0.250	0.199		
	taud		2.178	2.281	2.304	2.342	2.438	2.483	2.445	2.462	2.536	2.492	2.263	2.064		
	Ebn at Noon		171	229	260	269	278	282	272	266	257	220	162	128		
	Edh at Noon		12	20	28	32	32	31	31	29	22	17	11	8		
(44) All-Sky Solar Radiation	RadAvg		87	293	767	1246	1507	1702	1534	1197	784	348	112	52		
	RadStd		15	56	76	145	152	147	200	107	94	65	30	4		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page